



SEMEIA

MINISTÉRIO DO
DESENVOLVIMENTO
AGRÁRIO E
AGRICULTURA FAMILIAR

INTELIGÊNCIA ARTIFICIAL

SEMEIA System Engineering: Briefing & Roadmap

META-3 Coordination: Prof. Bruno
Automation Tech Lead: Antonio
AI Tech Lead: Mohamed

Outline

SEMEIA Overview

SEMEIA UI

SEMEIA UI Gateway

SEMEIA Backend AI System

SEMEIA Integration Gateway

SEMEIA Overview



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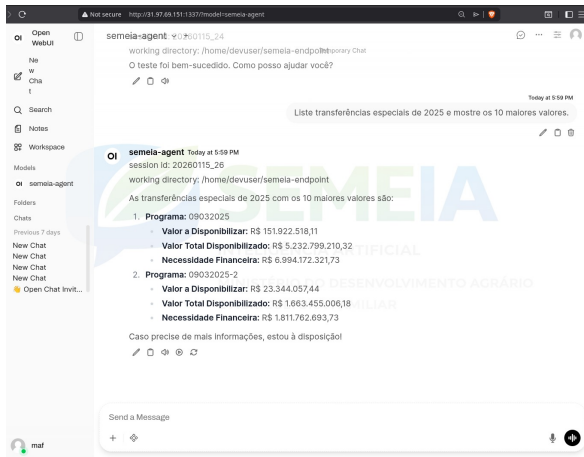
SEMEIA Integration Gateway

SEMEIA UI: Operational & Architectural Requirements

**SEMEIA UI enable
low-friction
prototyping while
aiming for for a
premium
domain-driven UX.**

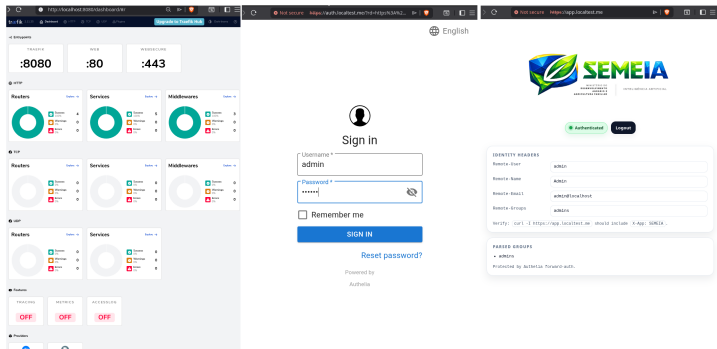
- **R1** – High-Velocity Iteration
- **R2** – Familiarity and Convenience
- **R3** – Bespoke MDA User Experience
- **R4** – Easy to Replace
- **R5** – Easy to Customize

SEMEIA UI: Current



openWebUI enables high-velocity iteration.

SEMEIA UI: Roadmap



Transitioning to a bespoke UI for a superior UX.

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SEMEIA UI Gateway: Operational & Architectural Requirements

SEMEIA UI Gateway should offer OpenAI-compatible API as well as custom endpoints.

- **R1** – OpenAI-comptiable API endpoints
- **R2** – Custom endpoints for specific features
- **R3** – Authentication and Authroization

SEMEIA UI Gateway – Current State



SEMEIA UI Gateway: Roadmap

**SEMEIA UI Gateway will
evolve to support UI
functions.**

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SEMEIA Backend AI System: Operational & Architectural Requirements

SEMEIA Backend AI is a compound AI system, not a model!

- **R1** – User-Agent Mapping: Who is this?
- **R2** – Context: What do we do?
- **R3** – Orchasteration: What should we do?
- **R4** – Governance: Is it allowed and safe?
- **R5** – Execution: How do we do it?
- **R6** – Quality: Did we do it correctly?

SEMEIA Backend AI – Current State



SEMEIA Backend AI: Roadmap

**Building User-specific
custom Agents.**

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SEMEIA Integration Gateway: Operational & Architectural Requirements

- **Configuration-based approach**

- **Developer Experience Matters**

- **Automate but Verify**

Gov APIs are not created equal, Act accordingly!

- **R1** – Handle well-documented gov APIs via Automation

- **R2** – Handle bad gov API via custom development

- **R3** – Enable configuration-based Developer Experience

- **R4** – GraphQL-based Gateway to aid Agent Usage

SEMEIA Integration Gateway – Current State



SEMEIA Integration Gateway: Roadmap

**End-to-end UX testing
of a real process.**

SEMEIA System Engineering: Briefing & Roadmap

Thank you

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